Moderated and Non-Linear Models.

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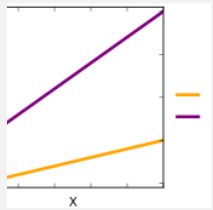
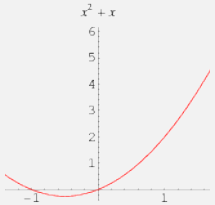
Definitions



Moderated: The effect of one variable changes as a function of another variable.



Nonlinear: The effect of one variable changes as a function of itself.

Moderated Models

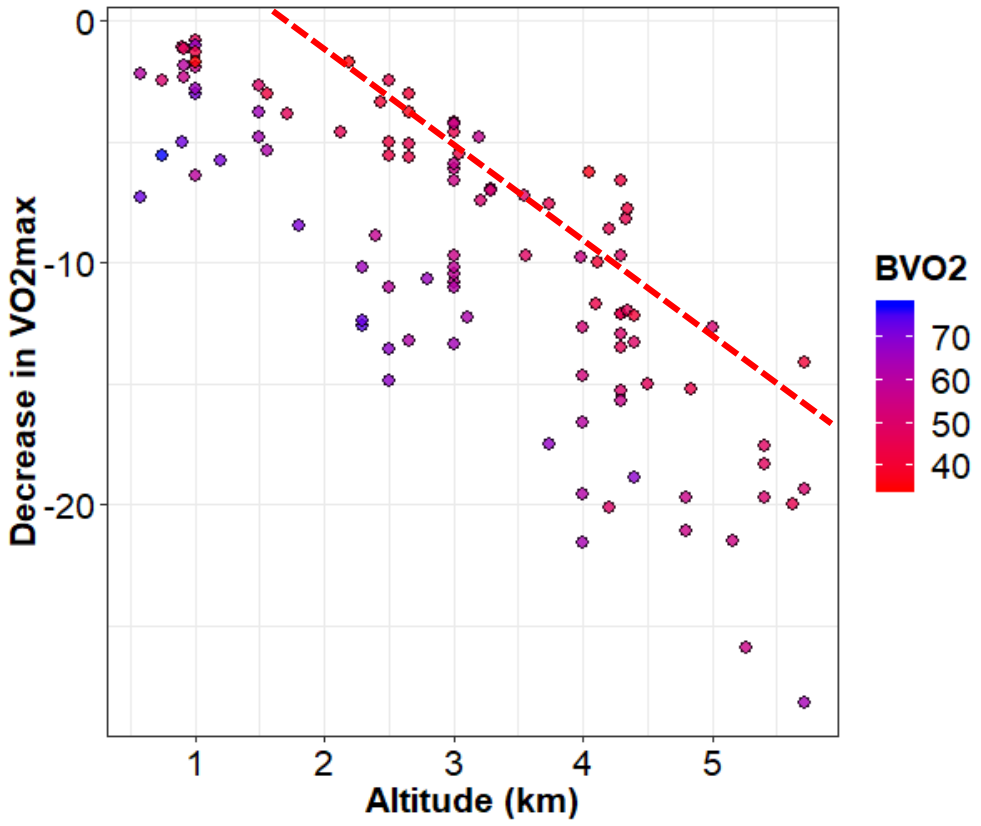
- Statistically, moderation refers to the **<u>interaction</u>** between two variables.

- For instance, in our altitude dataset, we were looking at the decrease in VO_{2max} as function of the altitude at which a test was conducted and an athlete's baseline VO_{2max}.

  - Unmoderated Model: $y_i = \beta_0 + \beta_1(ALT_i) + \beta_2(BVO2_i) + \epsilon_i$

Unmoderated (Additive) Model

Unmoderated Model: $y_i = 18.698 - 4.078(ALT_i) - 0.294(BVO2_i) + \epsilon_i$



$$y_i = [18.698 - 0.294(40)] - 4.078(ALT_i) + \epsilon_i$$

$$y_i = 6.938 - 4.078(ALT_i) + \epsilon_i$$

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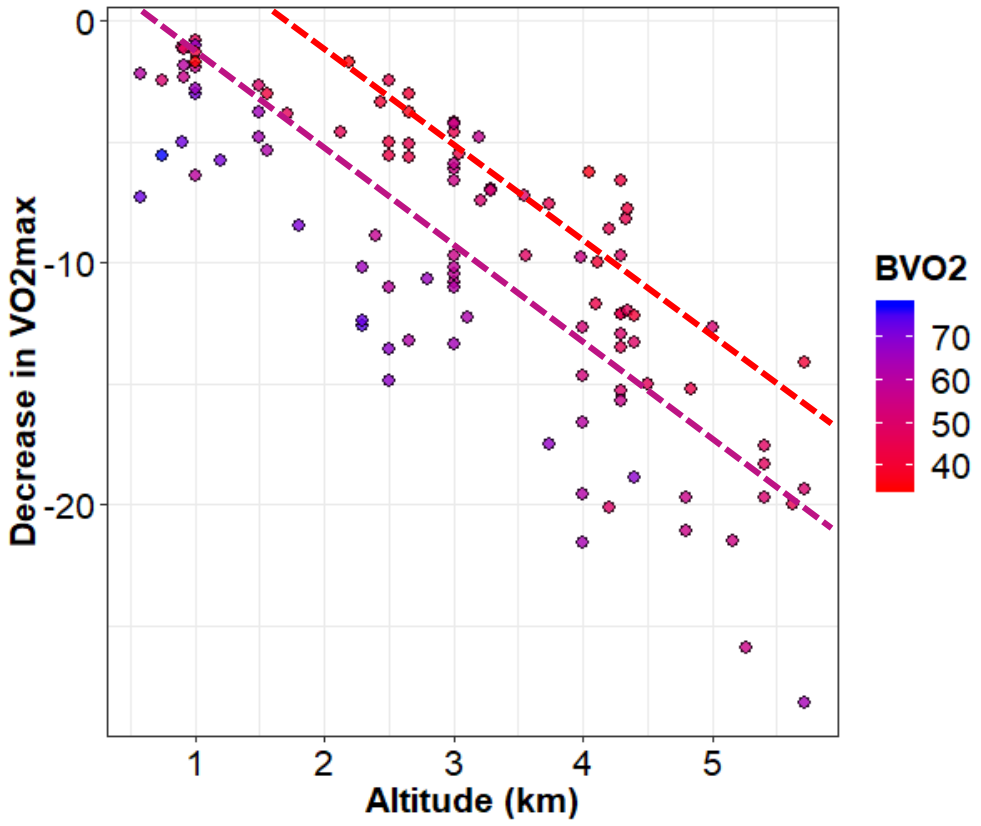

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$$y_i = [18.698 - 0.294(55)] - 4.078(ALT_i) + \epsilon_i$$

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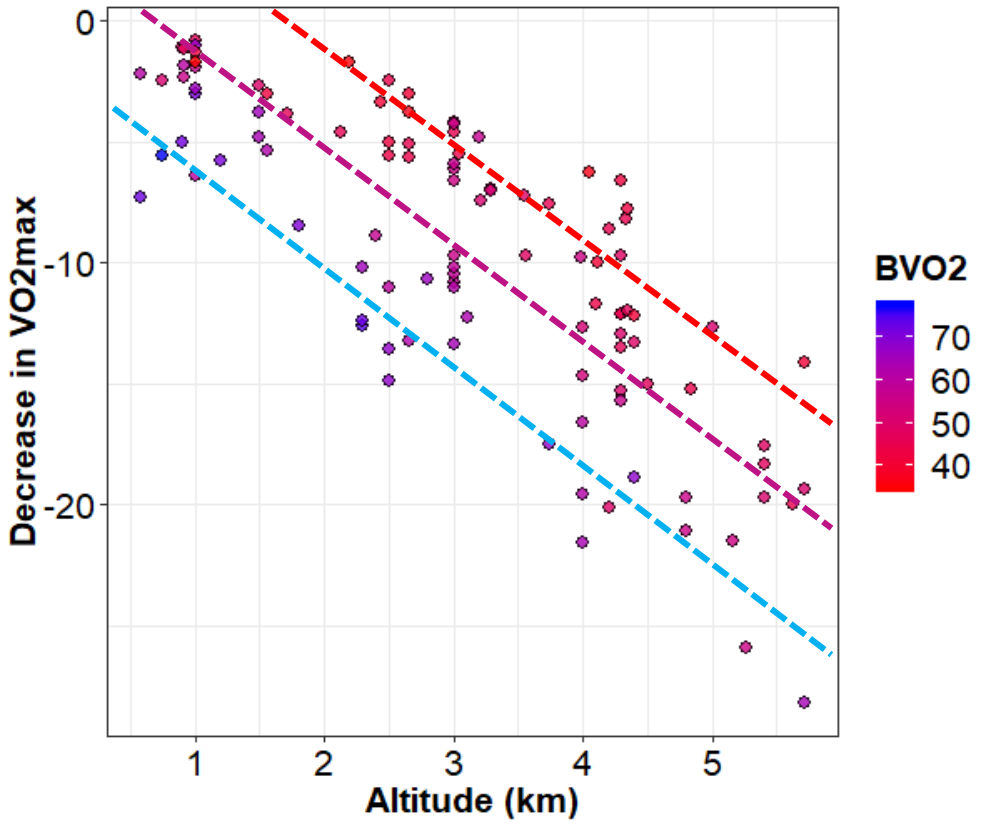

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$$y_i = [18.698 - 0.294(70)] - 4.078(ALT_i) + \epsilon_i$$
$$y_i = -1.882 - 4.078(ALT_i) + \epsilon_i$$

Moderated (Multiplicative) Models

- HOWEVER, we can also ask if the effect of altitude depends on a person's baseline fitness.
  - Some physiological data and theories suggest that fitter individuals actually experience a bigger decrease in VO_{2max} at altitude.
  - Their absolute VO_{2max} is still higher than less fit individuals, but their relative decline is greater.

- Interaction/Moderation: does the effect of altitude depend on fitness?
  - Moderated Model A: $y_i = \beta_0 + \beta_1(ALT_i) + \beta_2(BVO2_i) + \boldsymbol{\beta_3(ALT_i * BVO2_i)} + \epsilon_i$
  - Unmoderated Model C: $y_i = \beta_0 + \beta_1(ALT_i) + \beta_2(BVO2_i) + \epsilon_i$

How does a moderated model work?

- It probably isn't intuitive why multiplying two values together would allow them to "interact".

- Through some algebraic rearrangement, however, we can show why this works.

- What is the effect of Altitude?
  - Model A: $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$

How does a moderated model work?

- What is the effect of Altitude?
- Well, it depends on your BVO2:
  - Model A: $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$
  - Model A: $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(30) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \mathbf{30}) + \epsilon_i$
  - Model A: $y_i = [\beta_0 + \beta_2(30)] + [\boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \mathbf{30})] + \epsilon_i$
  - Model A: $y_i = [\beta_0 + \beta_2(30)] + [\boldsymbol{\beta_1} + \boldsymbol{\beta_3} * \mathbf{30})](\boldsymbol{ALT_i}) + \epsilon_i$

This is the 'simple slope' of altitude when BVO2 = 30!

How does a moderated model work?

- How does this change if your BVO2 increases?
  - Model A: $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$
  - Model A: $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(50) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \mathbf{50}) + \epsilon_i$
  - Model A: $y_i = [\beta_0 + \beta_2(50)] + [\boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \mathbf{50})] + \epsilon_i$
  - Model A: $y_i = [\beta_0 + \beta_2(50)] + [\boldsymbol{\beta_1} + \boldsymbol{\beta_3} * \mathbf{50})](\boldsymbol{ALT_i}) + \epsilon_i$

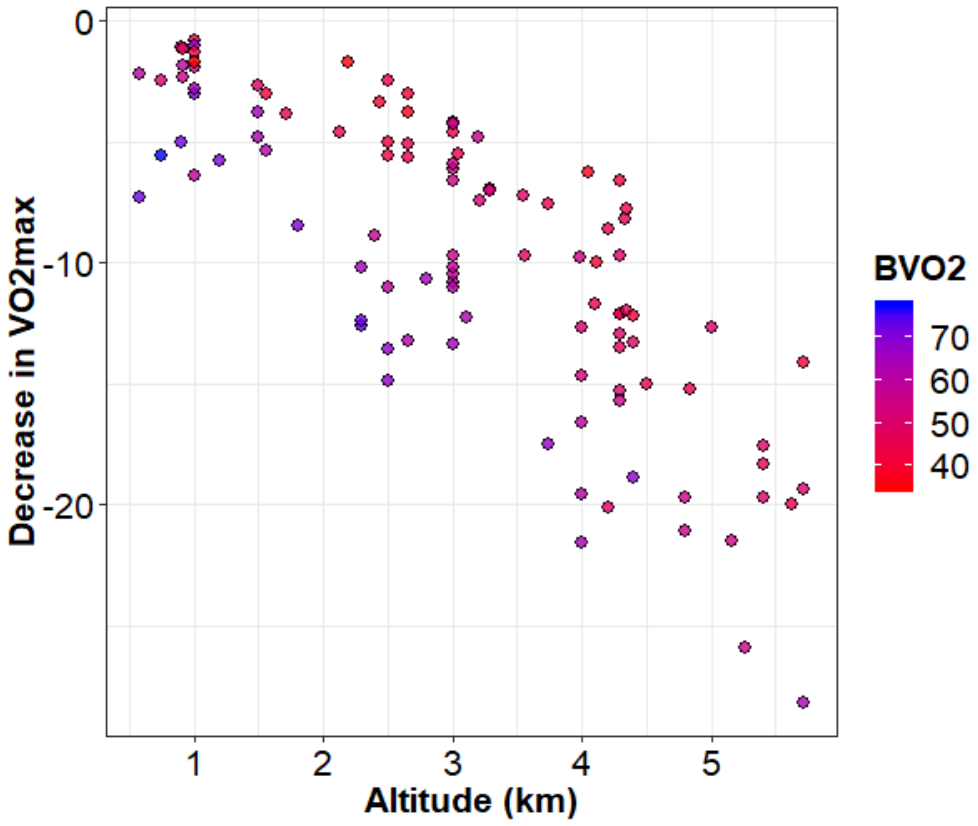
This is the 'simple slope' of altitude when BVO2 = 50!

Thus, in the moderated model, the effect of altitude changes with a person's BVO2.

Is there an interaction?

Using the ALTITUDE datafile, we can see how the VO_{2max} of 105 athletes changed as a function of the altitude at which their VO_{2max} was assessed, their baseline fitness, and the interaction of those terms.

- MODEL A: $y_i = \beta_0 + \beta_1(ALT_i) + \beta_2(BVO2_i) + \beta_3(ALT * BVO2_i) + \epsilon_i$
- MODEL C: $y_i = \beta_0 + \epsilon_i$



Is there an interaction?

```
lm(formula = decrease ~ alt_km * BVO2, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-7.2307 -1.4511 -0.1014  1.2200  6.9343 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  8.51588    3.01436   2.825 0.005697 ** 
alt_km      -0.11323    1.00029  -0.113 0.910097    
BVO2        -0.10874    0.05266  -2.065 0.041512 *  
alt_km:BVO2 -0.07448    0.01849  -4.027 0.000109 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.432 on 101 degrees of freedom
Multiple R-squared:  0.8488,	Adjusted R-squared:  0.8443 
F-statistic:   189 on 3 and 101 DF,  p-value: < 2.2e-16
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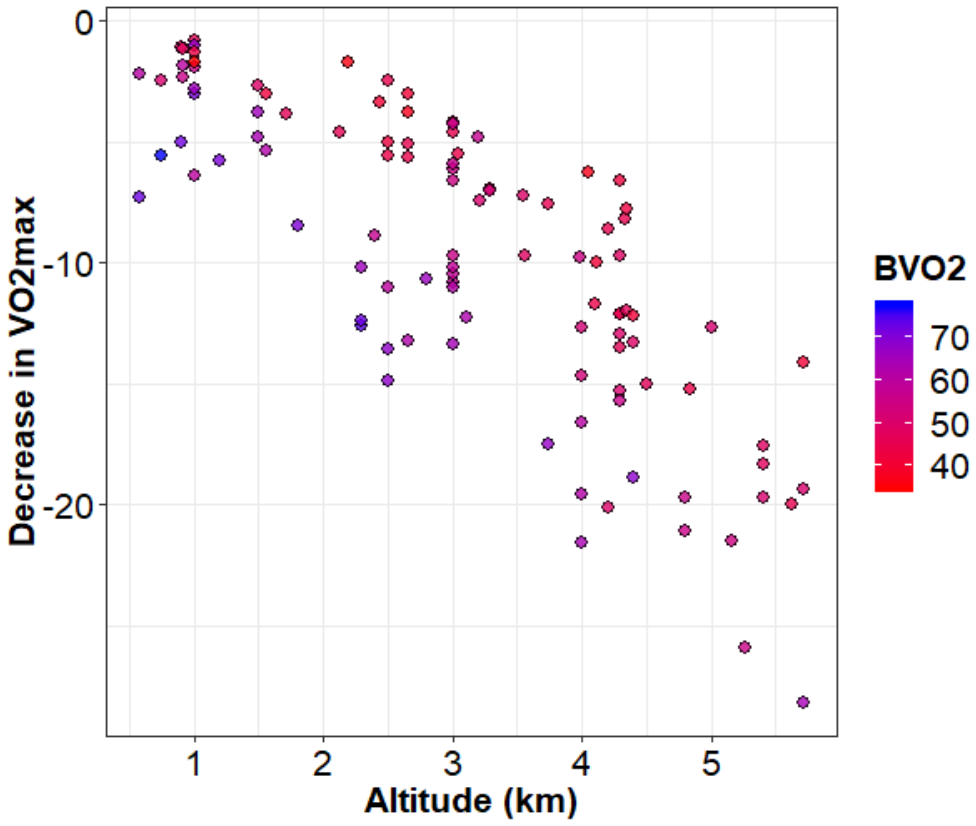

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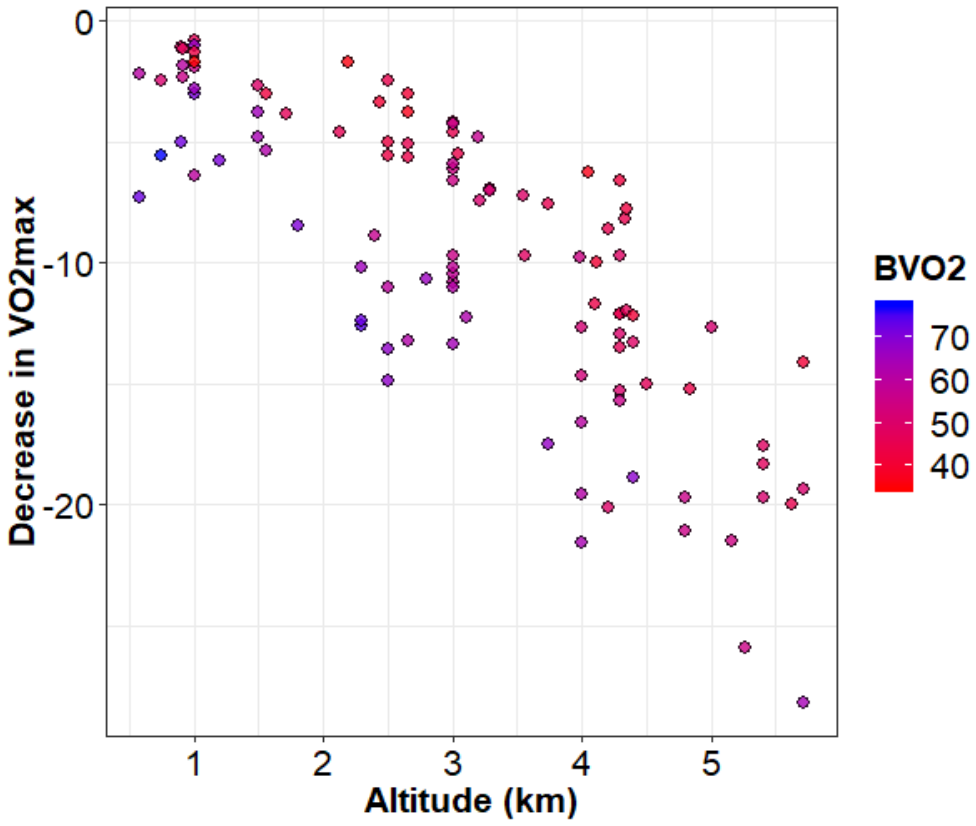
So what does our interaction mean?

The effect of altitude depends on your baseline fitness.

What is the effect of altitude for someone with a BVO2 = **40**?

- $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$
- $y_i = 8.516 - 0.113(ALT_i) - 0.109(BVO2_i) - 0.074\ (ALT_i * BVO2_i) + \epsilon_i$
- $y_i = [8.516 - 0.109(40)] + [\mathbf{-0.113 - 0.074 * 40}](\boldsymbol{ALT_i}) + \epsilon_i$
- $y_i = \mathbf{4.156} \mathbf{- 3.3073}(\boldsymbol{ALT_i}) + \epsilon_i$



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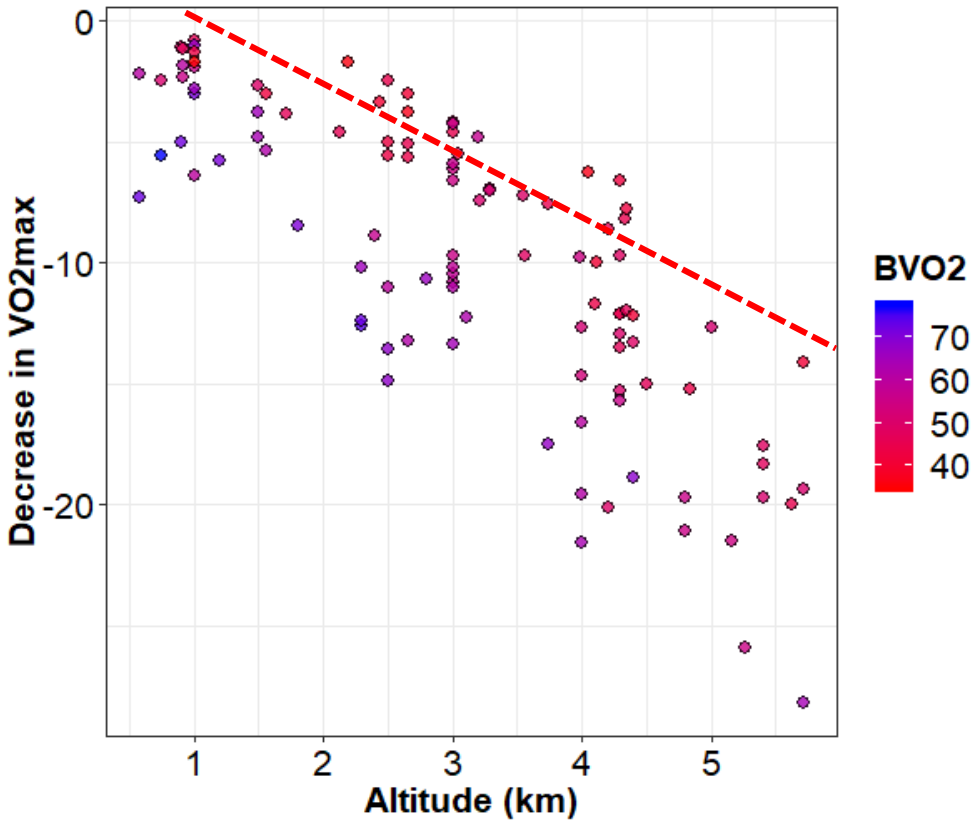

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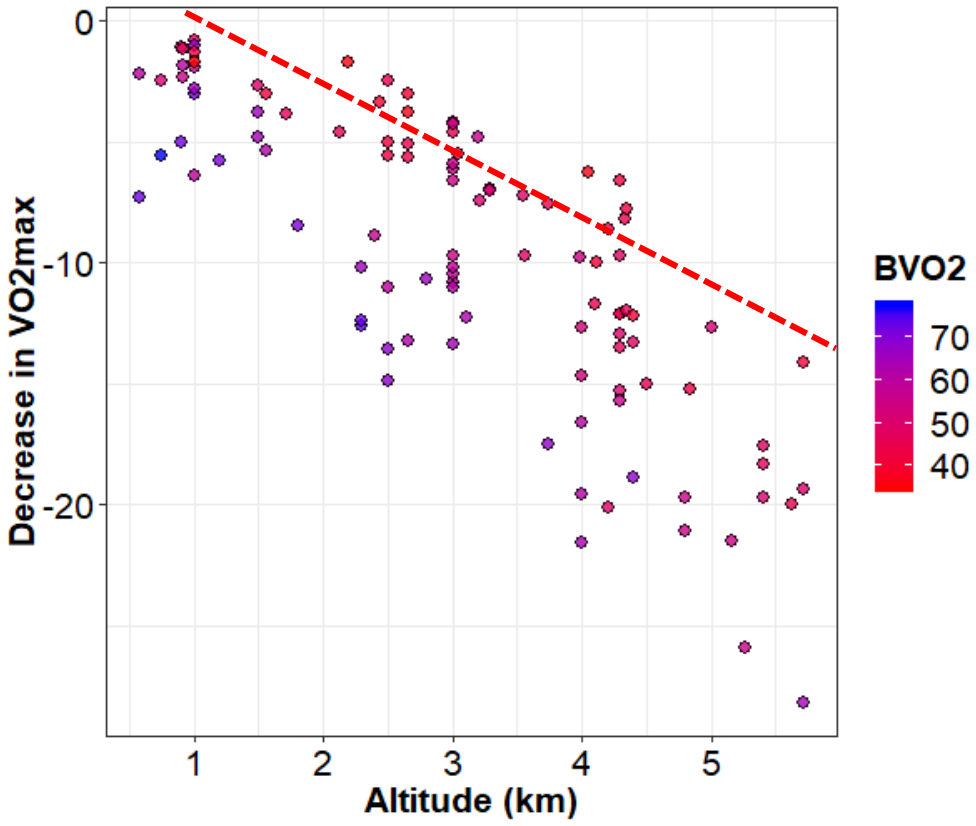
- $y_i = [8.516 - 0.109(55)] + [\mathbf{-0.113 - 0.074 * 55})](\boldsymbol{ALT_i}) + \epsilon_i$
- $y_i = \mathbf{2.521 - 4.183}(\boldsymbol{ALT_i}) + \epsilon_i$



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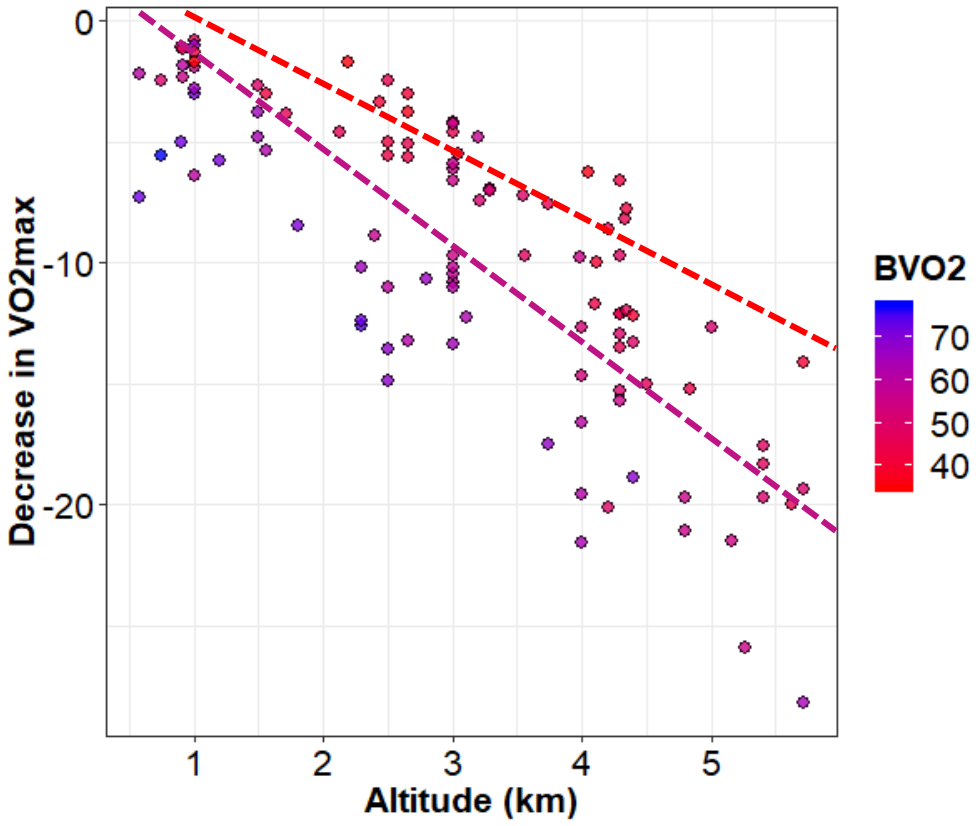
- $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$
- $y_i = 8.516 - 0.113(ALT_i) - 0.109(BVO2_i) - 0.074\ (ALT_i * BVO2_i) + \epsilon_i$
- $y_i = [8.516 - 0.109(55)] + [\mathbf{-0.113 - 0.074 * 55})](\boldsymbol{ALT_i}) + \epsilon_i$
- $y_i = \mathbf{2.521 - 4.183}(\boldsymbol{ALT_i}) + \epsilon_i$

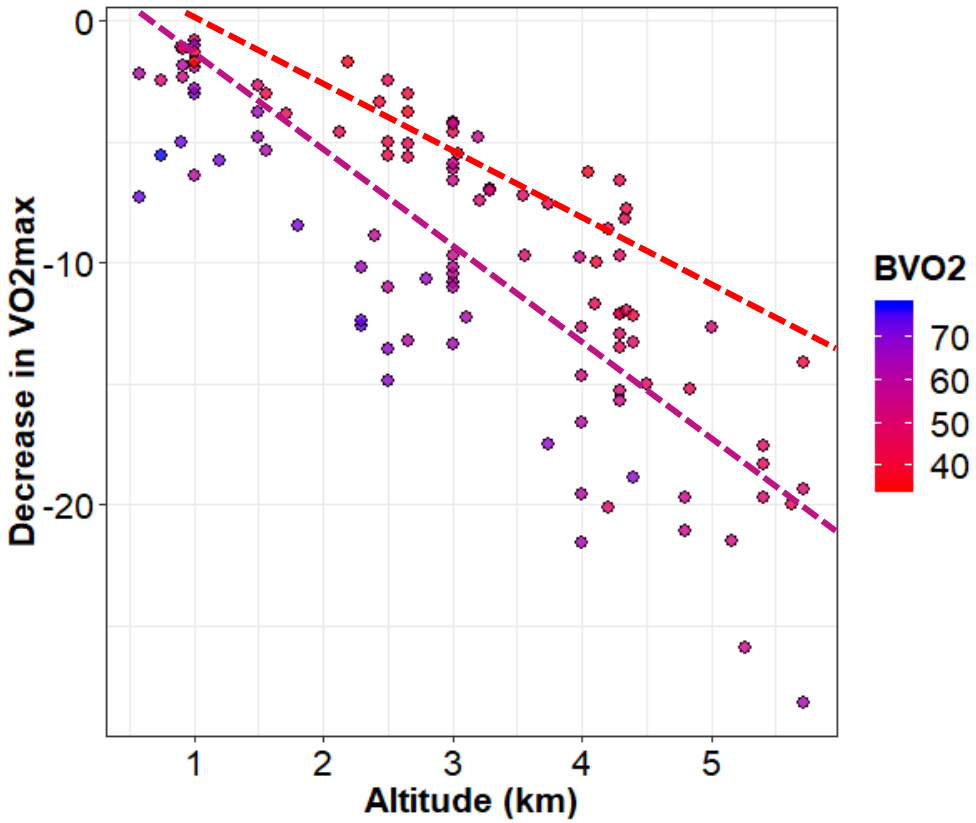


So what does our interaction mean?

The effect of altitude depends on your baseline fitness.

What is the effect of altitude for someone with a BVO2 = **70**?

- $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$
- $y_i = 8.516 - 0.113(ALT_i) - 0.109(BVO2_i) - 0.074\ (ALT_i * BVO2_i) + \epsilon_i$
- $y_i = [8.516 - 0.109(70)] + [\mathbf{-0.113 - 0.074 * 70}](\boldsymbol{ALT_i}) + \epsilon_i$
- $y_i = \mathbf{0.886} - \mathbf{5.293}(\boldsymbol{ALT_i}) + \epsilon_i$

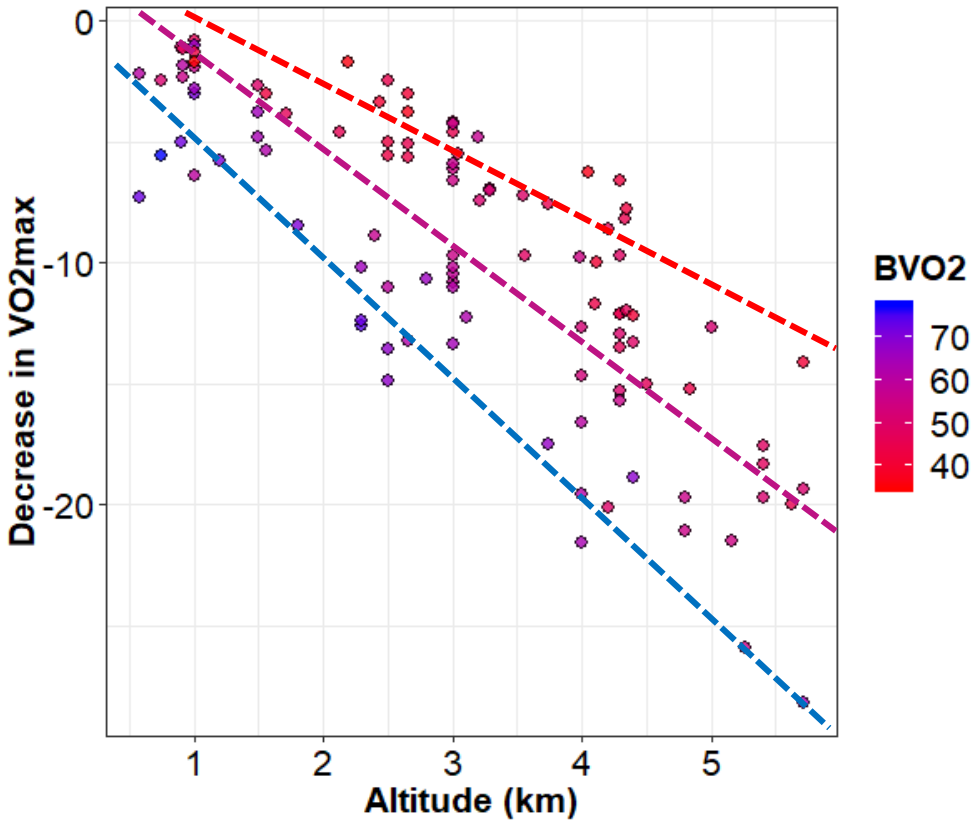


So what does our interaction mean?

The effect of altitude depends on your baseline fitness.

What is the effect of altitude for someone with a BVO2 = **70**?

- $y_i = \beta_0 + \boldsymbol{\beta_1}(\boldsymbol{ALT_i}) + \beta_2(BVO2_i) + \boldsymbol{\beta_3}(\boldsymbol{ALT_i} * \boldsymbol{BVO2_i}) + \epsilon_i$
- $y_i = 8.516 - 0.113(ALT_i) - 0.109(BVO2_i) - 0.074\,(ALT_i * BVO2_i) + \epsilon_i$
- $y_i = [8.516 - 0.109(70)] + [\mathbf{-0.113 - 0.074 * 70}](\boldsymbol{ALT_i}) + \epsilon_i$
- $y_i = \mathbf{0.886} - \mathbf{5.293}(\boldsymbol{ALT_i}) + \epsilon_i$

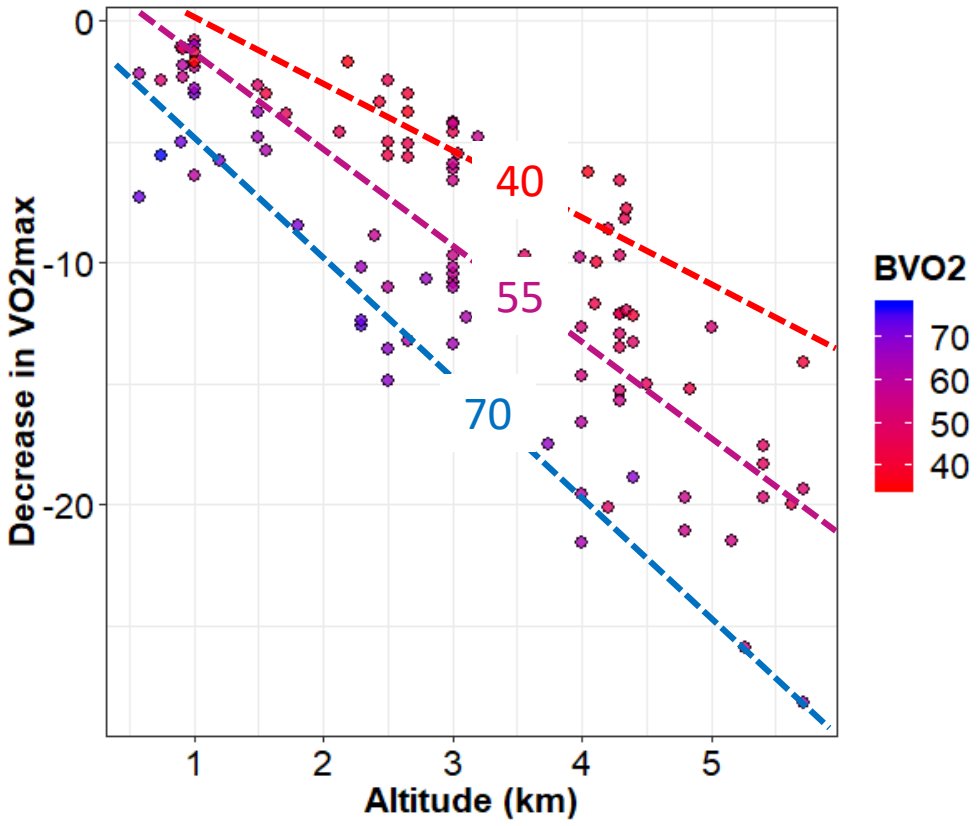


So what does our interaction mean?

The effect of altitude depends on your baseline fitness.

- $y_i = 8.516 - 0.113(ALT_i) - 0.109(BVO2_i) - 0.074\,(ALT_i * BVO2_i) + \epsilon_i$

- $y_i = \mathbf{4.156 - 3.3073(ALT_i)} + \epsilon_i$
- $y_i = \mathbf{2.521 - 4.183(ALT_i)} + \epsilon_i$
- $y_i = \mathbf{0.886 - 5.293(ALT_i)} + \epsilon_i$



ANOVA table for the full model

Source	SSE	Df	MSE	F_obs	Pr(>F_obs)
Intercept	47.21	1	47.21	7.981	0.006
Alt (km)	0.08	1	0.08	0.0128	0.910
BVO2	25.22	1	25.22	4.263	0.042
Alt * BVO2	95.95	1	95.95	16.221	<0.001
Residuals	597.44	101	5.92		
Total	765.90	104			